

LAB 1

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Q-1.PROGRAM TO CALCULATE SIMPLE INTEREST

CODE:-

```
#include<stdio.h>
int main() {
    float p,r,t,si;
    printf("enter the value of p,r,t : \n");
    scanf("%f%f%f", &p, &r, &t);
    si=p*r*t*0.01 ;
    printf("value of si : %f" , si);
    return 0;
}
```

Output:-

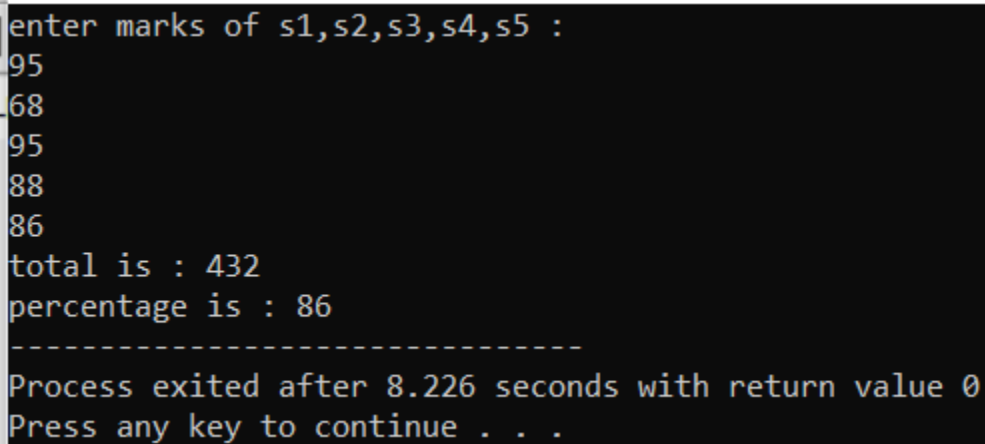
```
enter the value of p,r,t :
100
200
300
value of si : 60000.000000
-----
Process exited after 56.6 seconds with return value 0
Press any key to continue . . .
```

Q2)PROGRAM TO READ MARKS OF FIVE SUBJECT OF A STUDENT AND CALCULATE TOTAL AND PERCENTAGE

Code:-

```
#include<stdio.h>
main() {
    int s1,s2,s3,s4,s5,total,pr;
    printf("enter marks of s1,s2,s3,s4,s5 : \n");
    scanf("%d%d%d%d%d", &s1 , &s2 , &s3, &s4 , &s5);
    total=s1+s2+s3+s4+s5;
    printf("total is : %d\n" , total);
    printf("percentage is : %d" , total/5);
    return 0;
}
```

Output:-



```
enter marks of s1,s2,s3,s4,s5 :
95
68
95
88
86
total is : 432
percentage is : 86
-----
Process exited after 8.226 seconds with return value 0
Press any key to continue . . .
```

Q3)PROGRAM TO CALCULATE GROSS SALARY.

CODE:-

```
#include<stdio.h>
```

```
main()
```

```
{
```

```
    int previous_salary,after_salary;
```

```
    printf("enter the previous salary and the after salary \n");
```

```
    scanf("%d%d" , &previous_salary , &after_salary);
```

```
    printf("the gross salary is %d" , after_salary-previous_salary);
```

```
}
```

Output:-

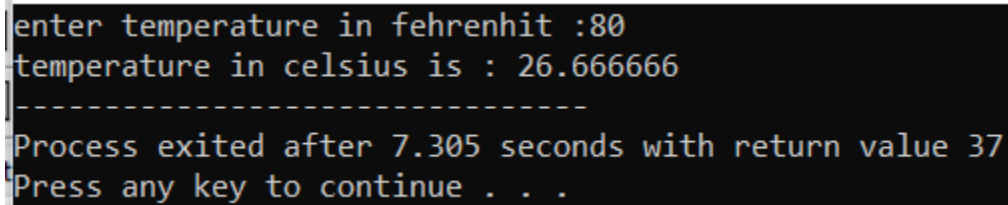
```
enter the previous salary and the after salary
50000
65000
the gross salary is 15000
-----
Process exited after 5.653 seconds with return value 25
Press any key to continue . . .
```

Q4)PROGRAM TO CONVERT TEMPERATURE FROM FAHRENHEIT TO CENTIGRADE DEGREES.

CODE:-

```
#include<stdio.h>
main()
{
    float tempf;
    printf("enter temperature in fehrenhit :");
    scanf("%f" , &tempf);
    printf("temperature in celsius is : %f" ,(tempf-32)*5/9);
}
```

Output:-

A screenshot of a terminal window showing the execution of the C program. The user enters '80' for the temperature in Fahrenheit. The program outputs 'temperature in celsius is : 26.666666'. Below this, a separator line of dashes is shown. The terminal then displays 'Process exited after 7.305 seconds with return value 37' and 'Press any key to continue . . .'.

```
enter temperature in fehrenhit :80
temperature in celsius is : 26.666666
-----
Process exited after 7.305 seconds with return value 37
Press any key to continue . . .
```

Q5)PROGRAM TO SWAP TOW VARIABLES USING THIRD.

CODE:-

```
#include<stdio.h>
```

```
main()
```

```
{
```

```
    int a,b,c;
```

```
    printf("enter the value of a,b\n");
```

```
    scanf("%d%d" , &a, &b);
```

```
    c=a;
```

```
    a=b;
```

```
    b=c;
```

```
    printf("after swap the value of a and b is : %d , %d" , a, b);
```

```
}
```

Output:-

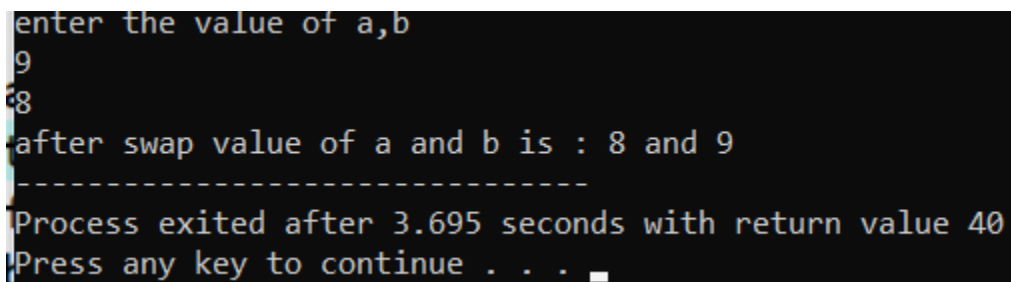
```
enter the value of a,b
5
6
after swap the value of a and b is : 6 , 5
-----
Process exited after 4.432 seconds with return value 42
Press any key to continue . . .
```

Q6)PROGRAM TO SWAP TWO VARIABLES WITHOUT USING THIRD VARIABLE.

CODE:-

```
#include<stdio.h>
main()
{
    int a,b;
    printf("enter the value of a,b\n");
    scanf("%d%d" , &a, &b);
    a=a+b;
    b=a-b;
    a=a-b;
    printf("after swap value of a and b is : %d and %d" , a , b);
}
```

Output:-



```
enter the value of a,b
9
8
after swap value of a and b is : 8 and 9
-----
Process exited after 3.695 seconds with return value 40
Press any key to continue . . .
```

Q7) PROGRAM TO CALCULATE AREA OF A TRIANGLE.

CODE:-

```
#include<stdio.h>
```

```
main()
```

```
{
```

```
    int base,altitude;
```

```
    printf("enter the value of base and altitude\n");
```

```
    scanf("%d%d" , &base , &altitude);
```

```
    printf("value of the area of triangle is : %f" , 0.5*base*altitude);
```

```
}
```

Output:-

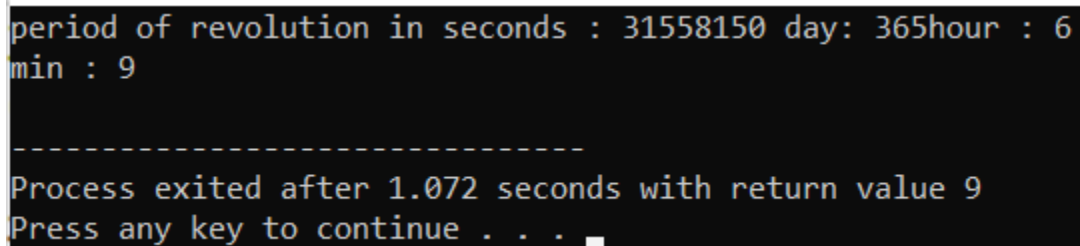
```
enter the value of base and altitude
5
6
value of the area of triangle is : 15.000000
-----
Process exited after 4.552 seconds with return value 44
Press any key to continue . . . ■
```

Q8)EARTH TAKES A PERIOD OF REVOLUTION OF 31558150 SECONDS
.WRITE A C PROGRAM TO CONVERT THIS INTO NUMBER OF DAY,HOURS
AND MINUTES

CODE:-

```
#include<stdio.h>
main()
{
    int period=31558150,hour,min,days;
    printf("period of revolution in seconds : %d " ,period);
    days= period/86400 ;
    printf("day: %d" , days);
    hour= period/3600 - days*24;
    printf("hour : %d \n" , hour);
    min= period/60 - hour*60 - days*24*60 ;
    printf("min : %d \n" , min);
}
```

Output:-

A screenshot of a terminal window showing the output of the C program. The output consists of three lines: "period of revolution in seconds : 31558150 day: 365hour : 6", "min : 9", and a separator line of dashes. Below the dashes, it says "Process exited after 1.072 seconds with return value 9" and "Press any key to continue . . .".

```
period of revolution in seconds : 31558150 day: 365hour : 6
min : 9
-----
Process exited after 1.072 seconds with return value 9
Press any key to continue . . .
```

Q9)PROGRAM TO READ TIME IN HR,MIN,SEC AND CONVERT IT INTO TOTAL SECOND.

CODE:-

```
#include<stdio.h>
main()
{
    int hours,min,sec,seconds;
    printf("enter the time in hours,minutes and second format :\n");
    scanf("%d%d%d", &hours,&min,&sec);
    seconds = hours*3600 + min*60 + sec ;
    printf("time in total second is : %d" , seconds);
}
```

Output:-

```
enter the time in hours,minutes and second format :
24
60
36
time in total second is : 90036
-----
Process exited after 11.39 seconds with return value 31
Press any key to continue . . .
```

Q10)WRITE A C PROGRAM TO CALCULATE THE CUT OFF MARK OF A STUDENT USING THE FORMULA. $CM = M/2 + P/2 + C/2 + E$ WHERE CM = Cut of f mark M = Marks in Mathematics out of 200 P = Marks in Physics out of 200 C = Marks in Chemistry out of 200
CODE:-

```
#include<stdio.h>
main()
{
    float m,p,c,e,cm;
    printf("enter the marks of mathematics,physics, chemistry and entrance examination\n");
    scanf("%f%f%f%f", &m , &p , &c ,&e);
    cm = m/2 + c/2 + p/2 + e ;
    printf("the cutoff marks of student is : %f" , cm);
}
```

Output:-

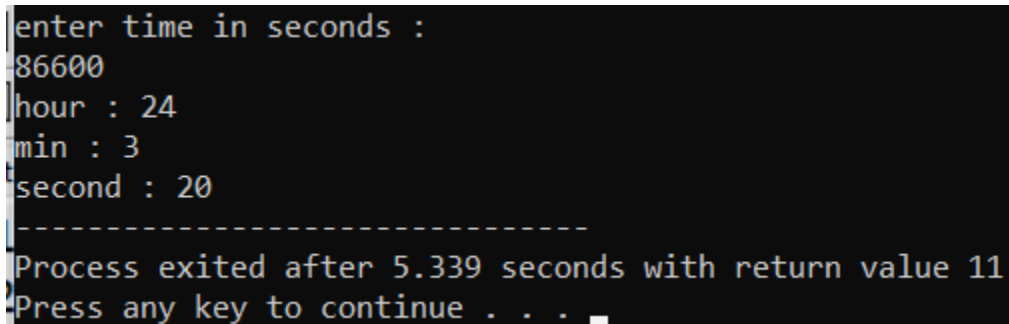
```
enter the marks of mathematics,physics, chemistry and entrance examination
185
196
158
96
the cutoff marks of student is : 365.500000
-----
Process exited after 10.64 seconds with return value 43
Press any key to continue . . .
```

Q11) PROGRAM TO READ TOTAL SECOND AND CONVERT IT INTO TIME.

CODE:-

```
#include<stdio.h>
main()
{
    int second,hour,min,sec;
    printf("enter time in seconds : \n");
    scanf("%d" , &second);
    hour= second/3600;
    printf("hour : %d \n" , hour);
    min= second/60 - hour*60 ;
    printf("min : %d \n" , min);
    sec= second - min*60 - hour*60*60 ;
    printf("second : %d" , sec);
}
```

Output:-



```
enter time in seconds :
86600
hour : 24
min : 3
second : 20
-----
Process exited after 5.339 seconds with return value 11
Press any key to continue . . .
```